Clinical Safety Case: AI-Assisted WHO Surgical Safety Checklist Verification Tool

DCB0129-aligned. Built as a self-directed portfolio project by Shaun Ukonu, RN. Not an approved NHS document; produced to demonstrate understanding of the standard.

1. Scope and Purpose

This Clinical Safety Case Report (CSCR) and accompanying Hazard Log have been produced in line with NHS England's DCB0129: Clinical Risk Management: its Application in the Manufacture of Health IT Systems (2018 edition — current at time of writing; a national review of DCB0129/DCB0160 is under consultation through 2026).

1.1 System description

An AI-assisted WHO surgical checklist verification tool that cross-references verbal checklist confirmations against a video/audio feed of the theatre, flagging unconfirmed items to the scrub team before incision.

1.2 Intended use

Used by the scrub team, circulating nurse, and lead surgeon during the WHO Surgical Safety Checklist (sign-in, time-out, and sign-out) within the theatre setting. The system listens to verbal checklist confirmations and cross-references them against a live video feed to verify that each required check (correct patient, correct site, correct procedure, instrument/swab counts) has actually been physically performed and not just verbally stated. It is intended as a secondary verification layer, running alongside — not replacing — the existing paper/EPR checklist process and standard theatre team communication.

1.3 Out of scope

The system does not make autonomous decisions, halt the procedure, or override clinical judgement it only flags unconfirmed items to the team for human review. It does not verify surgical technique, instrument sterility, or anaesthetic checks. It is not intended for use outside the theatre environment (e.g. ward-based procedures, endoscopy suites) without separate validation. It does not replace the statutory WHO checklist documentation requirement, which must still be completed in full regardless of system output. Performance has not been validated for theatres with non-standard layouts, poor lighting, or camera obstruction beyond the tested configuration.

2. Hazard Identification Method

Hazards below were identified using a combination of:

• Structured What-If Technique (SWIFT) — walking through each stage of the WHO checklist and asking 'what if the AI gets this wrong here?'

• Direct clinical experience — hazards drawn from real failure modes observed or plausible in scrub nurse practice.

• Fishbone analysis for HAZ-02 (automation bias) to separate human, process, and technology causes.

2.1 Risk management process

Hazards were identified through a structured walkthrough of the theatre checklist workflow (SWIFT — Structured What-If Technique), supplemented by Functional Failure Analysis of the system's core functions (speech recognition, video verification,

flagging logic). Each identified hazard was rated for severity and likelihood against the risk matrix in Section 2.2, controls were applied, and a residual risk score assigned. This process draws on the principles of ISO 14971 (risk management for medical devices) and follows the structure set out in DCB0129.

2.2 Risk matrix

Severity scale:

1 – Negligible: No harm / minor inconvenience

2 – Minor: Temporary harm, no intervention needed

3 – Moderate: Temporary harm requiring intervention

4 – Major: Permanent harm or major intervention required

5 – Catastrophic: Death or permanent severe harm

Likelihood scale:

1 – Rare: Very unlikely to occur

2 – Unlikely: Could occur but not expected

3 – Possible: Might occur at some point

4 – Likely: Will probably occur

5 – Almost certain: Expected to occur regularly

Risk = Severity × Likelihood

1–4 = LOW (acceptable, monitor)

5–9 = MEDIUM (mitigate where reasonably practicable)

10–14 = HIGH (mitigation required before release)

15–25 = UNACCEPTABLE (must not release without significant redesign)

2.3 Risk acceptability criteria

No hazard with a residual risk rating of HIGH or UNACCEPTABLE may proceed to release. Hazards rated MEDIUM must have a documented clinical justification for why the benefit of deployment outweighs the residual risk, reviewed and signed off by the Clinical Safety Officer. Hazards rated LOW require no further action beyond routine monitoring.

3. Hazard Log

ID	Hazard Description	Cause(s)	Effects / Harm	Initial Sev/Lik/Risk	Controls / Mitigations	Residual Sev/Lik/Risk	Status
HAZ-01	AI incorrectly reports checklist step (e.g., 'Sign Out') as complete when it was skipped or rushed.	Speech recognition misclassifies partial/background conversation as the checklist phrase; model overconfidence in noisy theatre environment.		Cat 3 / Likely / High	1) Require explicit verbal confirmation phrase from lead clinician, not passive detection. 2) Visual on-screen prompt requiring manual tick alongside AI flag. 3) Audit log of every AI-detected completion, reviewable post-op.	Cat 3 / Possible / Medium	Open
HAZ-02	Automation bias: scrub/circulating staff stop independently verifying checklist steps because they trust the AI's tracking.	Repeated correct AI performance builds unwarranted trust; workload pressure incentivises reliance on the tool.		Cat 4 / Likely / High	1) Mandatory training on automation bias at induction (links to DCB0129 training requirement).2) Tool explicitly framed in UI as 'decision support', never 'decision maker'.3) Periodic deliberate audit where AI tracking is disabled to test staff independent compliance.	Cat 4 / Unlikely / Medium	Open
HAZ-03	System continues verifying against the original planned procedure after an emergency intraoperative conversion (e.g. laparoscopic cholecystectomy converted to open due to unexpected	The system's checklist parameters are set at sign-in based on the listed procedure and have no mechanism to detect or accept a real-time change in procedure type once surgery is underway; no team member manually updates the system during an emergency, since		Cat 5 / Possible / Unacceptable	Mandatory manual "procedure change" trigger built into the theatre's existing emergency conversion protocol, prompting a designated team member (not the operating surgeon) to update the system at the point of conversion; system displays a persistent, unmissable visual banner stating "VERIFICATION PAUSED — PROCEDURE CHANGE NOT CONFIRMED"	Cat 5/ Unlikley / high	Open

	bleeding), rather than recognising the procedure has changed	attention is on the clinical event, not the software			if conversion isn't acknowledged, rather than continuing to run silently against stale parameters; system defaults to manual-verification-only mode (no automated pass/fail signal) whenever a conversion trigger is unconfirmed		
HAZ-04	The team treats a "no discrepancies" output as proof of safety and reduces vigilance on manual swab, instrument and site checks (automation bias).	Reliable performance over time builds complacency; time pressure; the system's output feels more objective than a human count.	A retained item or wrong-site error that the system also missed goes undetected because the human safety net has weakened.	Cat 5 / Likely / Unacceptable	The system never displays "verified" or "safe", only "no discrepancies detected". Manual counts remain mandatory under existing policy. Training covers system limits. Periodic simulation drills with deliberately seeded discrepancies test whether the team still catches them.	Cat 5 / Unlikely / High	Open
HAZ-05	Detection accuracy drops in a theatre whose layout, lighting, camera position or equipment differs from the training data.	Training data drawn predominantly from a limited set of theatre configurations.	Unconfirmed items are missed (false negatives) or false alarms flood the team. Both erode the effectiveness of the secondary check.	Cat 4 / Likely / Unacceptable	Site-specific acceptance testing before go-live in each theatre. A documented supported-configuration envelope (lighting, camera position, layout). The system warns or declines to run when conditions fall outside that envelope. Staged rollout, starting in shadow mode with no live alerts.	Cat 4 / Unlikely / Medium	Open
HAZ-06	Performance degrades after go-live through a vendor model update, changes in PPE or instrument sets, or gradual drift, and nobody notices.	No ongoing performance monitoring; updates pushed without re-validation.	Detection quietly worsens, so the team relies on a tool that is less accurate than when it was validated.	Cat 4 / Possible / High	hange control requiring re-validation before any model update. Ongoing monitoring against agreed accuracy thresholds, using sampled cases reviewed by humans. Model version displayed and logged on every list. Rollback capability to the last validated version.	Cat 4 / Unlikely / Medium	Open
HAZ-07	An integration fault with the theatre	Message delay or error between systems; list	The system verifies the	Cat 5 / Possible / Unacceptable	Patient identity confirmed against at least two	Cat 5 / Rare / Medium	Open

	management or patient record system loads the wrong patient or procedure, or a list that has since been reordered.	changes not propagated; interface mapping fault.	checklist against the wrong patient's details, creating a false assurance of correct patient and procedure.		identifiers by the team at sign-in, before the system loads the checklist. Patient name and procedure displayed on screen and manually acknowledged. Interface messages timestamped and validated, with automatic fallback to manual mode on failure. Interface responsibilities agreed with the deploying organisation.		

4. Clinical Safety Case Report

4.1 System and scope

This report covers [Product Name], an AI-assisted tool that supports the WHO Surgical Safety Checklist (sign-in, time-out and sign-out) in the operating theatre. It cross-references verbal confirmations against a live video feed and flags unconfirmed items to the team for human review. It operates as a secondary verification layer alongside the existing checklist process and does not make autonomous decisions. The full system description, intended use and exclusions are set out in Section 1.

4.2 Summary of risk management process followed

Risk management followed the process defined in Section 2. Hazards were identified through a SWIFT walkthrough of the theatre checklist workflow, supplemented by Functional Failure Analysis of the system's core functions. Each hazard was rated for severity and likelihood against the risk matrix in Section 2.2. Controls were then identified, and a residual risk was assigned and checked against the acceptability criteria in Section 2.3. Hazards are recorded in the Hazard Log (Section 3). Hazard identification should be treated as ongoing, and the log will be reviewed whenever the system, its use or its environment changes.

4.3 Summary of hazards and residual risk

ID	Hazard	Initial rating	Residual rating
HAZ-01	Unconfirmed item not flagged due to poor video feed	Cat 5 / Possible / Unacceptable	Cat 5 / Rare / Medium
HAZ-02	Speech recognition mishears site/side confirmation	Cat 5 / Possible / Unacceptable	Cat 5 / Rare / Medium
HAZ-03	System verifies against original procedure after emergency conversion	Cat 5 / Possible / Unacceptable	Cat 5 / Unlikely / High
HAZ-04	Team over-trusts system and relaxes manual checks	Cat 5 / Likely / Unacceptable	Cat 5 / Unlikely / High
HAZ-05	Model fails to generalise to a different theatre	Cat 4 / Likely / Unacceptable	Cat 4 / Unlikely / Medium
HAZ-06	Model drift or unvalidated update degrades performance	Cat 4 / Possible / High	Cat 4 / Unlikely / Medium

HAZ-07	Interface fault loads wrong patient or stale list	Cat 5 / Possible / Unacceptable	Cat 5 / Rare / Medium

Of seven hazards, none has a residual risk rated Unacceptable. Two (HAZ-03, HAZ-04) are rated High. Five are rated Medium. None is rated Low.

4.4 Outstanding risks and risk-benefit statement

HAZ-03 (emergency conversion): Residual risk is High because the control depends on a team member updating the system during a clinical emergency, and human reliability under pressure cannot be fully engineered out. The system fails safe: if the conversion is not acknowledged, it displays a persistent banner and reverts to manual-verification-only mode rather than continuing to produce automated pass/fail signals. The existing manual checklist and count processes are unaffected. On this basis the residual risk is considered acceptable, subject to simulated-drill verification before go-live and monthly audit of unconfirmed-conversion events.

HAZ-04 (automation bias): Residual risk is High because complacency is a behavioural risk that training and interface design can reduce but not eliminate. The system never displays "verified" or "safe", only "no discrepancies detected". Manual counts remain mandatory, and seeded-discrepancy drills test whether the team's vigilance holds. Because the tool adds a layer to existing safeguards and removes none, the net effect on patient safety is considered positive, provided the quarterly drill results show no decline in manual detection.

Medium residual risks (HAZ-01, 02, 05, 06, 07): Each has layered controls and a fail-safe fallback to manual verification. The residual likelihood of each is Rare or Unlikely, and the benefit of an additional, objective check is considered to outweigh the remaining risk.

Per Section 2.3, High residual risks proceed only with additional controls and sign-off by the Clinical Safety Officer and the deploying organisation's senior clinical lead. This approval would be required before any real deployment.

4.5 Overall safety statement

Based on the risk management process described above, [Product Name] is considered safe for use as a secondary verification aid during the WHO Surgical Safety Checklist in a defined theatre environment, subject to the controls in Section 3 being implemented and verified, site-specific acceptance testing being completed, and the High residual risks being formally accepted as set out in Section 4.4. This statement applies to the system as defined in Section 1 and should be reviewed on any change to the system, its intended use, or its deployment environment.

4.6 Sign-off

Clinical Safety Officer: Shaun Ukonu
Role / registration: Registered Nurse, Training exercise, not an issued safety case.
Date: 13/10/26

Reflection

Building this showed me how much of clinical safety is about imagining how a system fails in real use, not just how it's meant to work. The hazard I'm proudest of is HAZ-03, the emergency conversion. Anyone reading the standard could write about camera failures or speech errors. But I've been in theatre when a case changes mid-list, and I know nobody is looking at a screen at that point: the whole room is focused on the patient. That's why I designed the control to fail safe, so the system pauses visibly rather than relying on someone to update it.

What surprised me was that the honest answer for some hazards is a High residual risk, and the skill is justifying it clearly rather than hiding it.